

NUMBER SYSTEM

OBJECTIVES:

1. To distinguish the different number systems.
2. To convert a number system to another number system.

Number Systems

Decimal, Binary, Octal and Hexadecimal
Numeric Systems.

Conversions among the different
systems

Decimal Numbering systems

Base: 10

Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Decimal Numbering systems

Base: 10

Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Binary Numbering systems

- Base: 2
- Digits: 0, 1
- binary number: 110101_2

positional powers of 2:	2^5	2^4	2^3	2^2	2^1	2^0
decimal positional value:	32	16	8	4	2	1
binary number:	1	1	0	1	0	1

Binary to Decimal Conversion

- To convert to base 10, add all the values where a one digit occurs.

Ex: 110101_2

positional powers of 2: 2^5 2^4 2^3 2^2 2^1 2^0

decimal positional value: 32 16 8 4 2 1

binary number: 1 1 0 1 0 1

$$32 + 16 + 4 + 1 = 53_{10}$$

Ex: 101011_2

positional powers of 2: 2^5 2^4 2^3 2^2 2^1 2^0

decimal positional value:

binary number:

Ex: 101011_2

positional powers of 2: 2^5 2^4 2^3 2^2 2^1 2^0

decimal positional value: 32 16 8 4 2 1

binary number: 1 0 1 0 1 1

$$32 + 8 + 2 + 1 = 43_{10}$$

- *The Division Method.* Divide by 2 until you reach zero, and then collect the remainders in reverse.

Ex 1: $56_{10} = 111000_2$

$2 \overline{) 56}$ Rem:

$2 \overline{) 28}$ 0

$2 \overline{) 14}$ 0

$2 \overline{) 7}$ 0

$2 \overline{) 3}$ 1

$2 \overline{) 1}$ 1

0 1

Ex 2: $35_{10} =$

2)

Rem:

2)

2)

2)

2)

2)

Answer: $35_{10} =$

2

Character Representation

Ex: Find the binary ASCII and decimal ASCII values for the '&' character.

Rightmost Four Bits	Leftmost Three Bits							
	<u>000</u>	<u>001</u>	<u>010</u>	<u>011</u>	<u>100</u>	<u>101</u>	<u>110</u>	<u>111</u>
0000	NUL	DLE	Space	0	@	P	,	p
0001	SOH	DC1	!	1	A	Q	a	q
0010	STX	DC2	"	2	B	R	b	r
0011	ETX	DC3	#	3	C	S	c	s
0100	EOT	DC4	\$	4	D	T	d	t
0101	ENQ	NAK	%	5	E	U	e	u
0110	ACK	SYN	<u>&</u>	6	F	V	f	v
0111	BEL	ETB	'	7	G	W	g	w
1000	BS	CAN	(	8	H	X	h	x
1001	HT	EM	)	9	I	Y	i	y
1010	LF	SUB	*	:	J	Z	j	z
1011	VT	ESC	+	;	K	[	k	{
1100	FF	FS	,	<	L	\	l	
1101	CR	GS	-	=	M	]	m	}
1110	SO	RS	.	>	N	^	n	~
1111	SI	US	/	?	O	_	o	DEL

Character Representation

ASCII Table

From the chart:

'&' = 0100110 (binary ASCII value)

Convert the binary value to decimal:

$$0100110_2 = 32 + 4 + 2 = 38_{10}$$

Therefore:

'&' = 38 (decimal ASCII value)

Octal Numbering systems

- Base: 8
- Digits: 0, 1, 2, 3, 4, 5, 6, 7
- Octal number: 1246_8

powers of :	8^4	8^3	8^2	8^1	8^0	
decimal value:	4096	512	64	8	1	
Octal number:			1	2	4	6

Octal to Decimal Conversion

- To convert to base 10, beginning with the rightmost digit multiply each n th digit by $8^{(n-1)}$, and add all of the results together.

Ex: 1246_8

positional powers of 8: 8^3 8^2 8^1 8^0

decimal positional value: 512 64 8 1

Octal number: 1 2 4 6

$$512 + 128 + 32 + 6 = 678_{10}$$

Ex: 10352_8

positional powers of 8: 8^4 8^3 8^2 8^1 8^0

decimal positional value:

Octal number:

- *The Division Method.* Divide by 8 until you reach zero, and then collect the remainders in reverse.

Ex 1: $4330_{10} = 10352_8$

$$8 \overline{) 4330}$$

Rem:

$$8 \overline{) 541}$$

2

$$8 \overline{) 67}$$

5

$$8 \overline{) 8}$$

3

$$8 \overline{) 1}$$

0

0

1

Ex 2: $810_{10} =$

8) 810 Rem:

8)

8)

8)

Answer: $810_{10} =$ 8

Hexadecimal Numbering systems

- Base: 16
- Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Hexadecimal number: $1F4_{16}$

powers of :	16^4	16^3	16^2	16^1	16^0
decimal value:	65536	4096	256	16	1
Hexadecimal number:			1	F	4

<u>Four-bit Group</u>	<u>Decimal Digit</u>	<u>Hexadecimal Digit</u>
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

Hexa to Decimal Conversion

- To convert to base 10, beginning with the rightmost digit multiply each n th digit by $16^{(n-1)}$, and add all of the results together.

Ex: $1F4_{16}$

positional powers of 16: 16^3 16^2 16^1 16^0

decimal positional value: 4096 256 16 1

Hexadecimal number: 1 F 4

$$256 + 240 + 4 = 500_{10}$$

Ex: $7E_{16}$

positional powers of 16: 16^3 16^2 16^1 16^0

decimal positional value:

Hexa number:

- *The Division Method.* Divide by 16 until you reach zero, and then collect the remainders in reverse.

Ex 1: $126_{10} = 7E_{16}$

$$16 \overline{) 126} \quad \text{Rem:}$$

$$16 \overline{) 7} \quad 14=E$$

$$0 \quad 7$$

Ex 2: $810_{10} =$

16) 810 Rem:

16)

16)

Answer: $810_{10} =$ 16

Binary to Octal Conversion

- Since the maximum value represented in 3 bit is equal to:

$$2^3 - 1 = 7$$

- i.e. using 3 bits we can represent values from 0 –7 which are the digits of the Octal numbering system.
- Thus, three binary digits can be converted to one octal digit and visa versa.

<u>Three-bit Group</u>	<u>Decimal Digit</u>	<u>Octal Digit</u>
000	0	0
001	1	1
010	2	2
011	3	3
100	4	4
101	5	5
110	6	6
111	7	7

Ex :

Convert $742_8 =$ 2

$$7 = 111$$

$$4 = 100$$

$$2 = 010$$

$$742_8 = 111\ 100\ 010_2$$

Ex :

Convert $10100110_2 =$ $\quad\quad\quad_8$

$$110 = 6$$

$$100 = 4$$

$$010 = 2 \quad (\text{pad empty digits with 0})$$

$$10100110_2 = 246_8$$

<u>Four-bit Group</u>	<u>Decimal Digit</u>	<u>Hexadecimal Digit</u>
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

Ex :

Convert $3D9_{16} =$ 2

$$3 = 0011$$

$$D = 1101$$

$$9 = 1001$$

$$3D9_{16} = 0011\ 1101\ 1001_2$$

Ex :

Convert $10100110_2 =$ $\quad\quad\quad_{16}$

$$0110 = 6$$

$$1010 = A$$

$$10100110_2 = A6_{16}$$

- To convert between Octal to Hexadecimal numbering systems and visa versa convert from one system to binary first then convert from binary to the new numbering system

Ex :

Convert $E8A_{16} =$ 8

1110 1000 1010₂

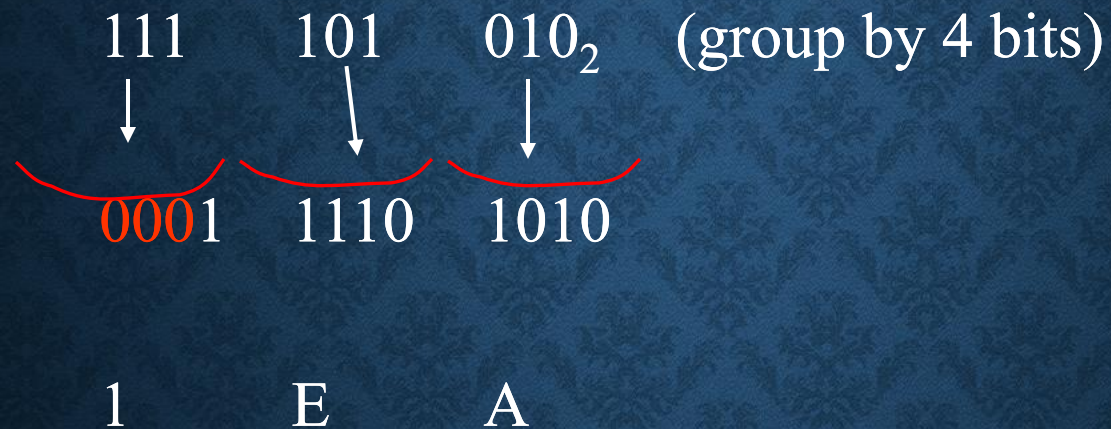
111 010 001 010 (group by 3 bits)

7 2 1 2

$E8A_{16} = 7217_8$

Ex :

Convert $752_8 =$ $_{16}$



$$752_8 = 1EA_{16}$$

THANK YOU FOR LISTENING!! ☺